ICT702 Data Visualisation

Data Visualisation and Design

Module 3

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USC AUSTRALIA

USC

I would like to acknowledge the traditional custodians of the land we are meeting on,

The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

Part 3

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Minimizing Eye Travel

We can minimize the cognitive load needed to view and interpret data visualization by:

- using preattentive attributes
- using Gestalt principles
- increasing the data-ink ratio
- **minimizing eye travel**

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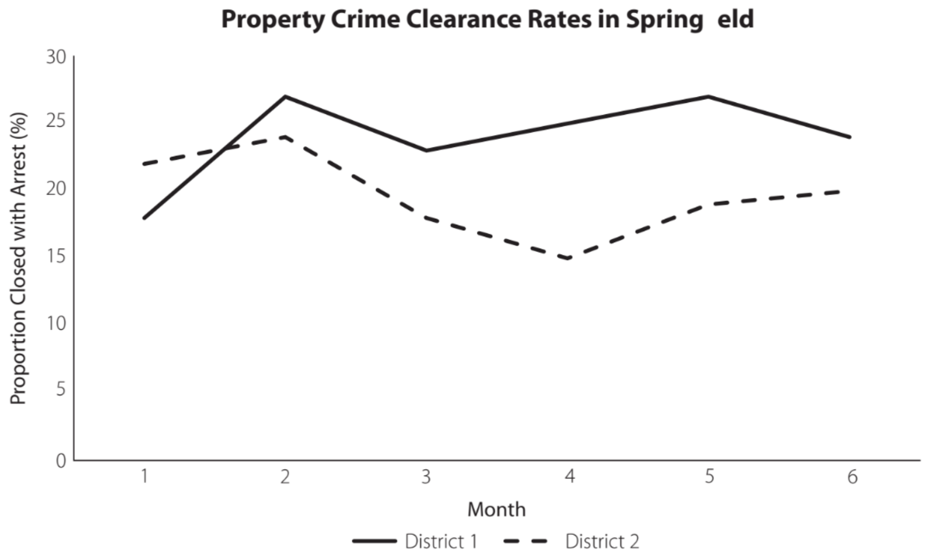
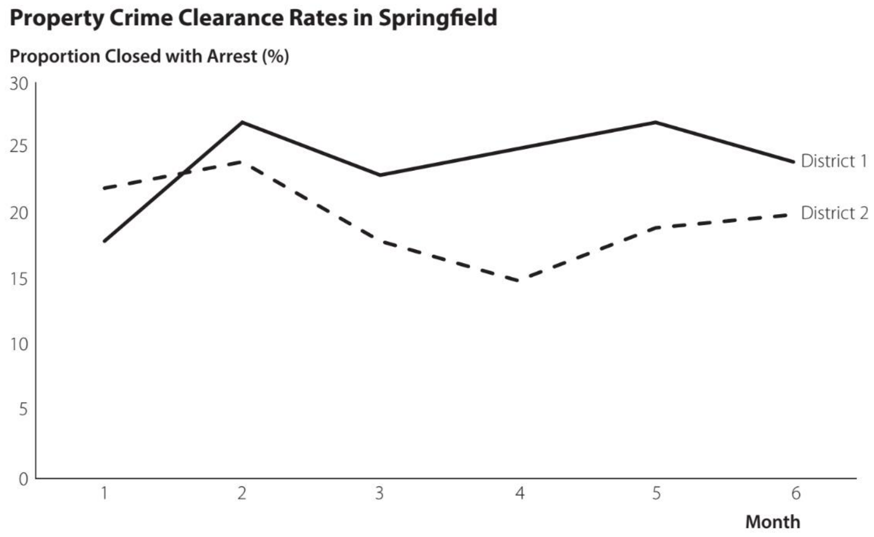
Minimizing Eye Travel in a Chart

A default line chart created in Excel for property crime clearance rates



Improved line chart for property crime clearance rates – Minimized eye travel



Required eye travel is minimized by:

- Aligning the chart title and vertical-axis title with the vertical axis
- Moving the line labels adjacent and to the right of the chart lines
- Aligning the horizontal-axis label with the line labels at the end of the horizontal axis.

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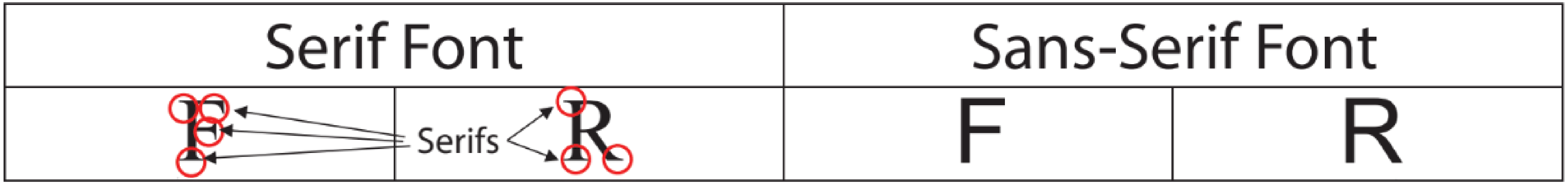
Choosing a Font for Text



Serif Font

- Preferred for text in printed work
- Includes end-of-stroke features
- Less legible at small sizes
- Common fonts include:
  - Times, Times New Roman, and Courier

Sans-Serif Font

- Preferred for text in data visualization
- Does not Includes end-of-stroke features
- More legible at small sizes
- Common fonts include:
  - Arial, Calibri, Myriad Pro, and Verdana

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Suggestions for Choice of Font for Text

- The choice of the font in data visualization should be consistent and restrained.
- Most data visualizations should use a single type of font for text to minimize cognitive load.
- Horizontal- and vertical-axis titles should have the same font type and size.
- The use of different font sizes, **bold**, *italic*, and colour enhancements, help differentiate between features and direct audience attention.

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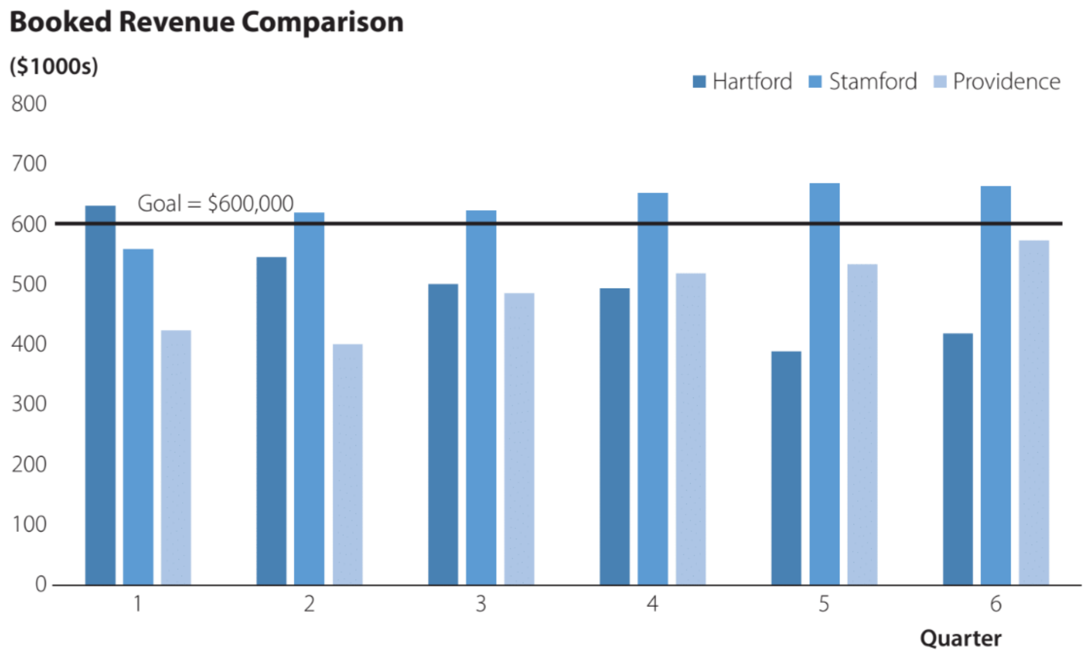
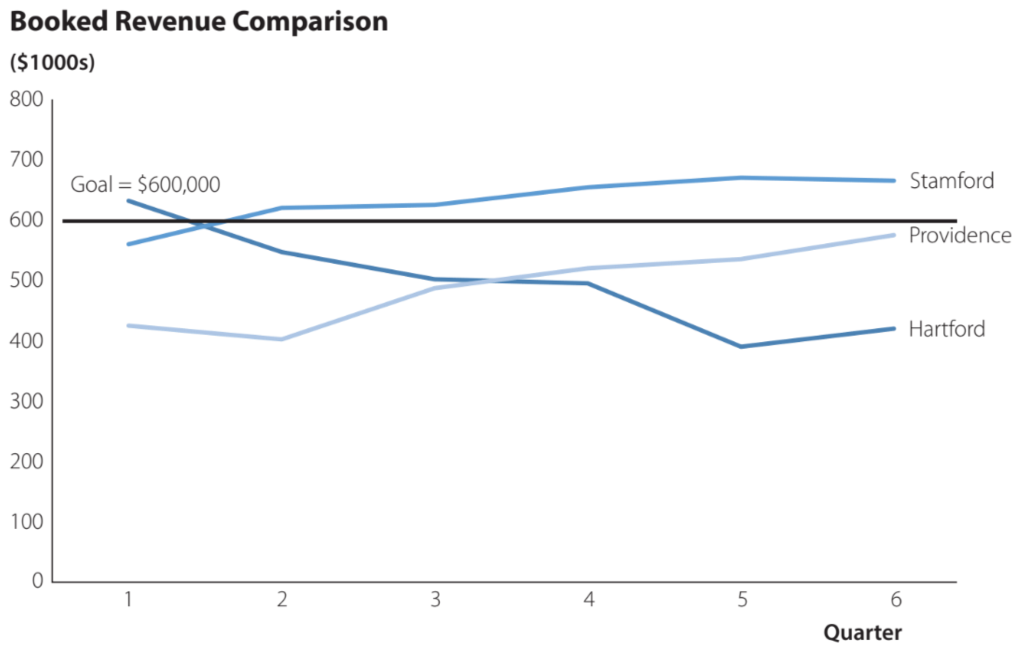
The Wrong Type of Visualization

A cluttered clustered column chart for the booked revenue data comparison



An improved line chart for the booked revenue data comparison



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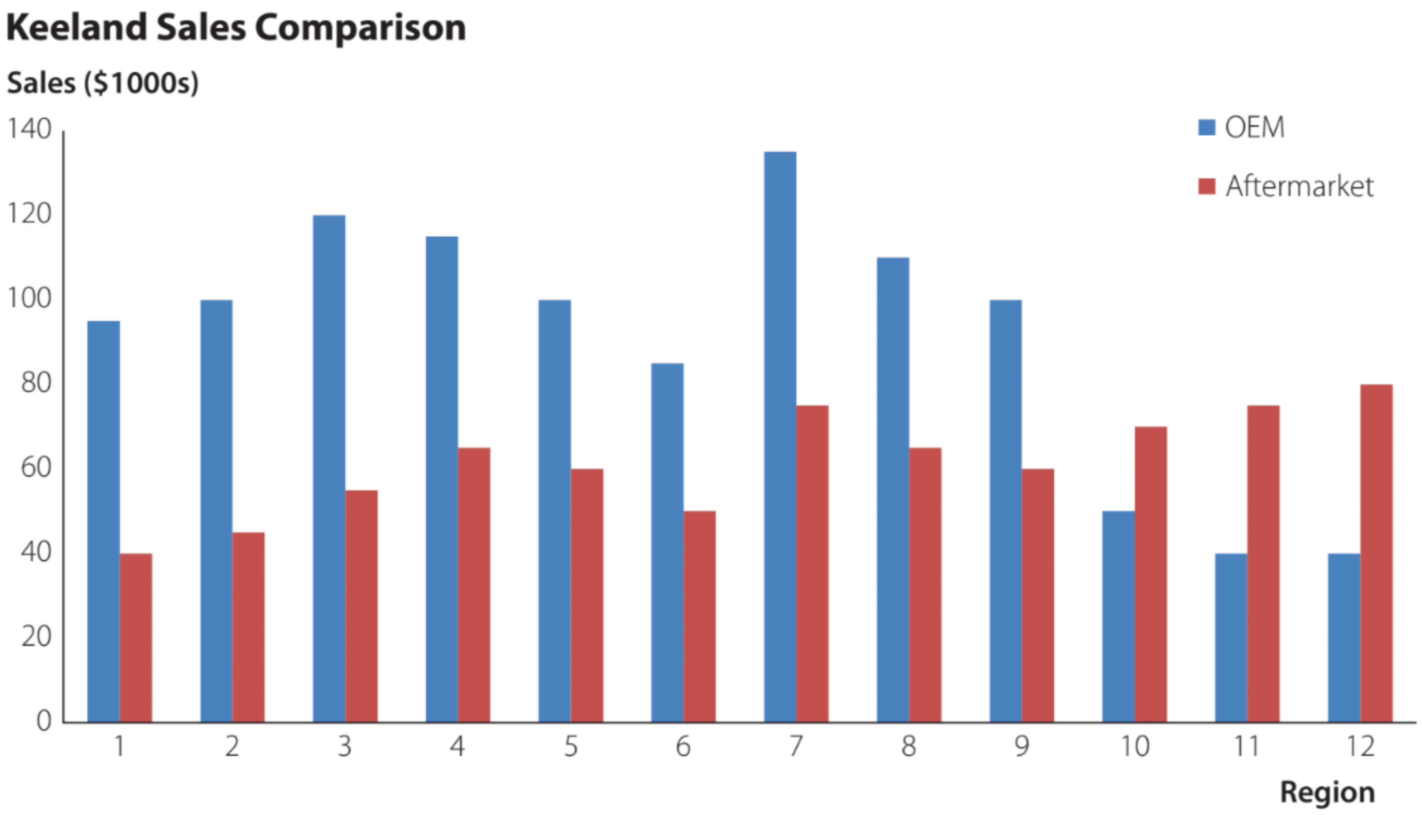
Displaying Excessive Information

A cluttered clustered column chart showing OEM and aftermarket sales for several regions.



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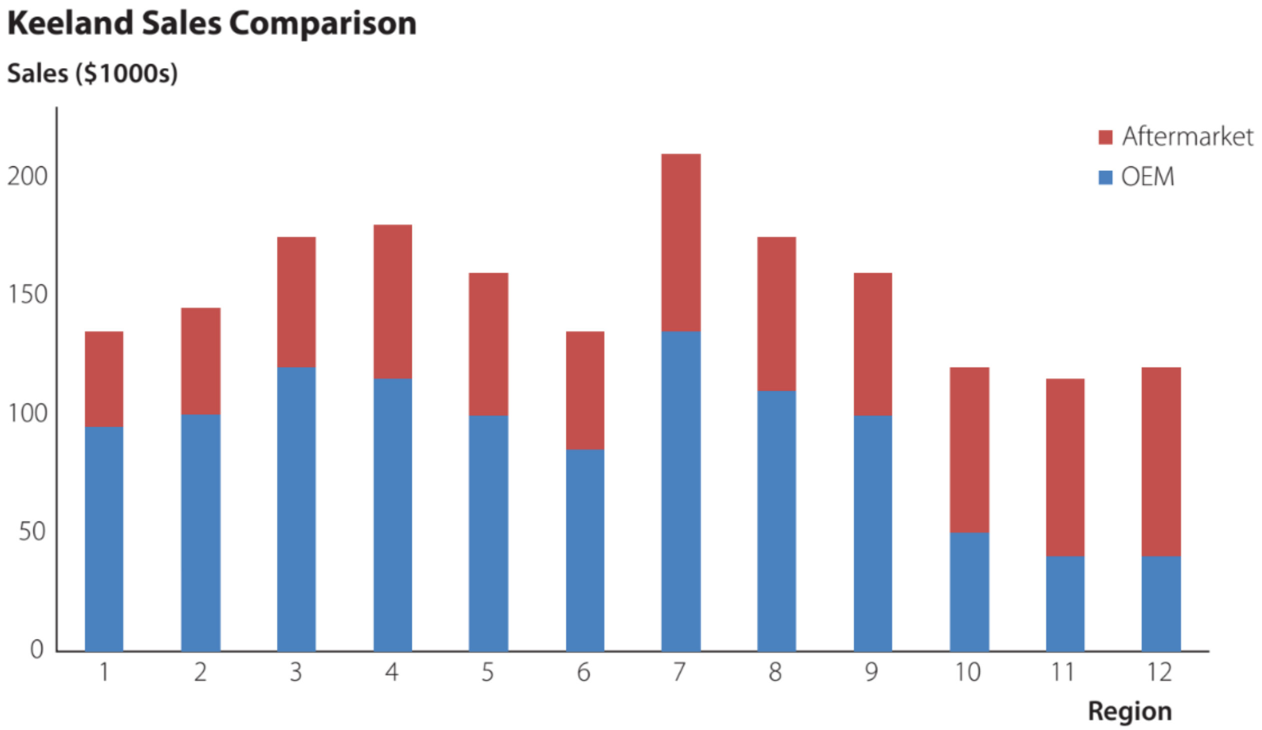
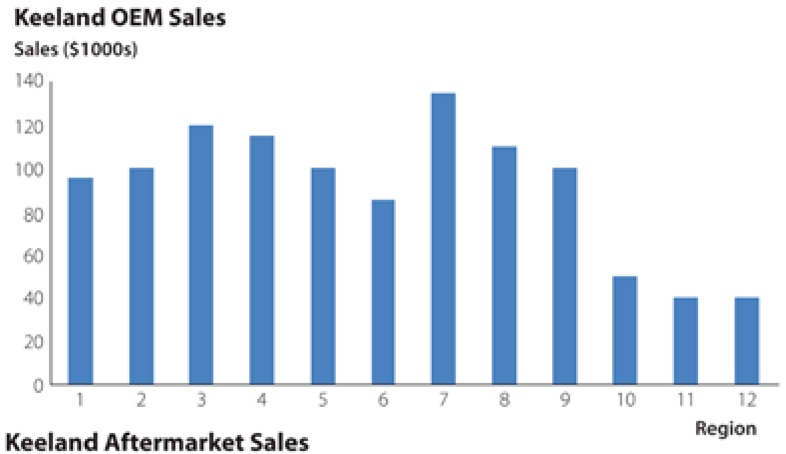
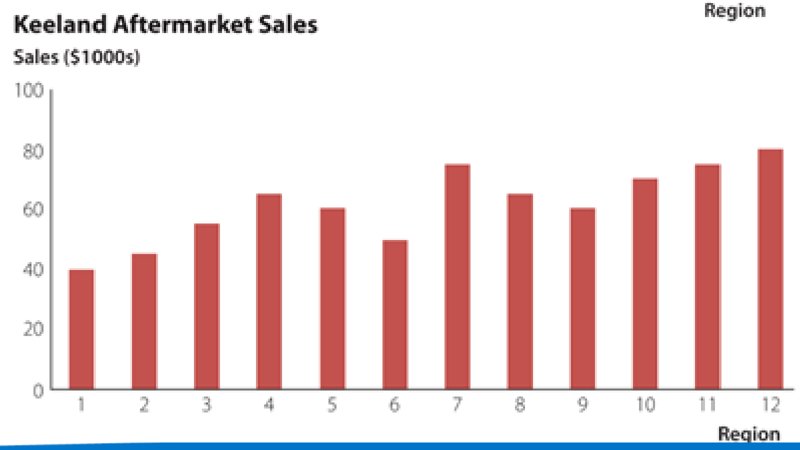
Displaying Excessive Information – Two Solutions

A stacked clustered column chart showing OEM and aftermarket sales for several regions.



Improved column charts showing OEM and aftermarket sales separately





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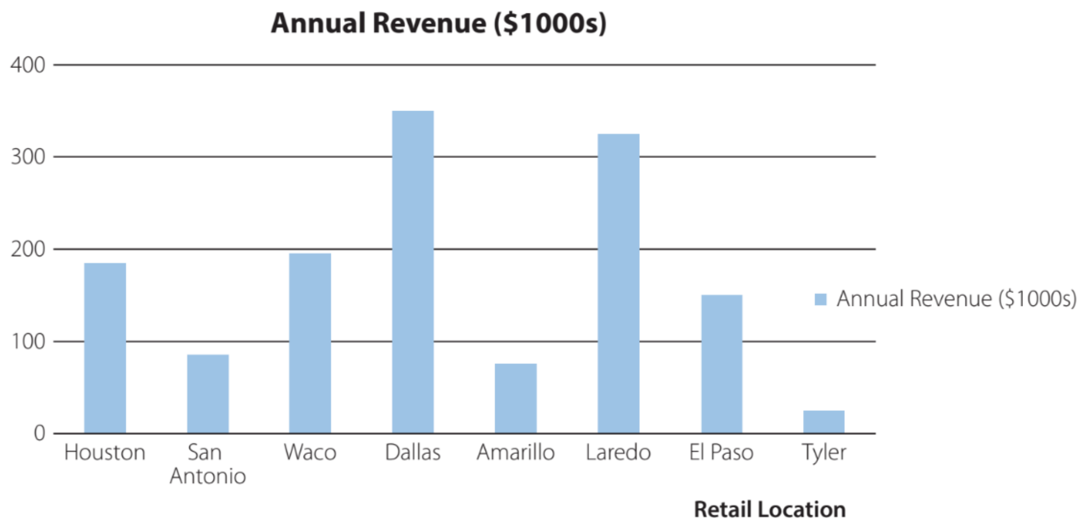
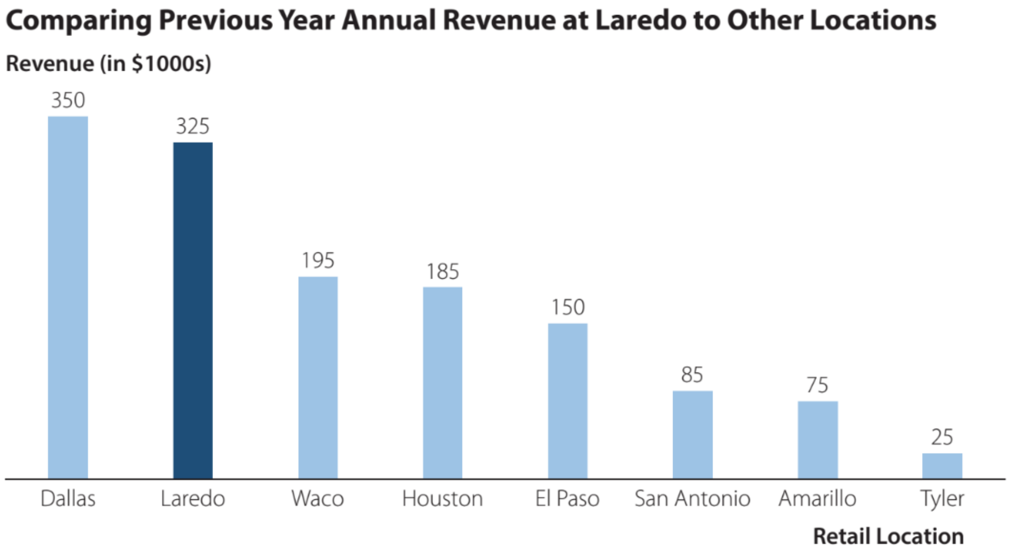
Using Excel Default Settings for Charts

Default Column Chart Comparing Annual Revenue at Eight Retail Locations



Improved Column Chart Comparing Annual Revenue at Eight Retail Locations



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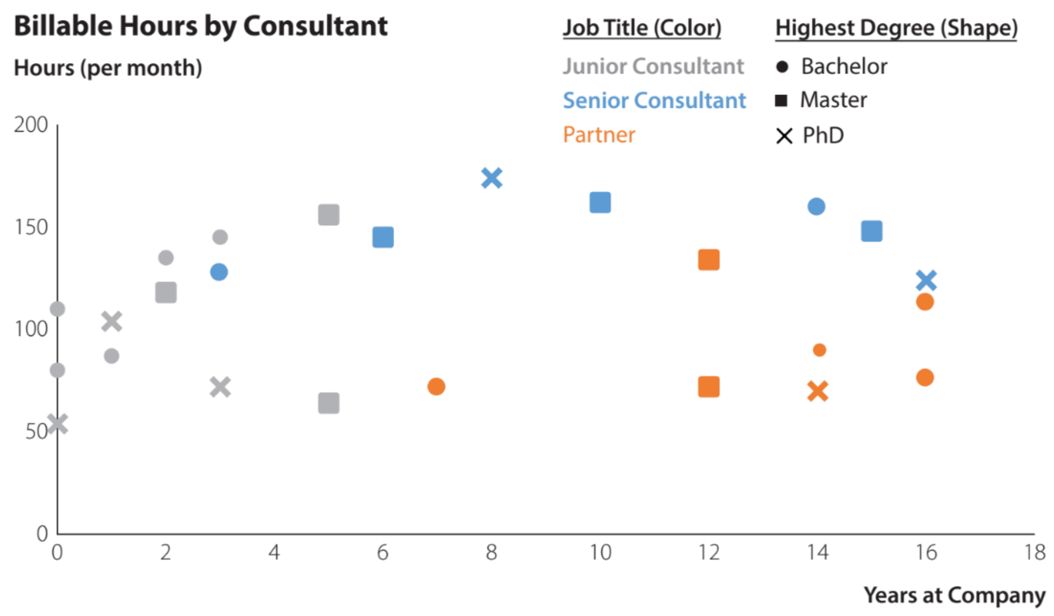
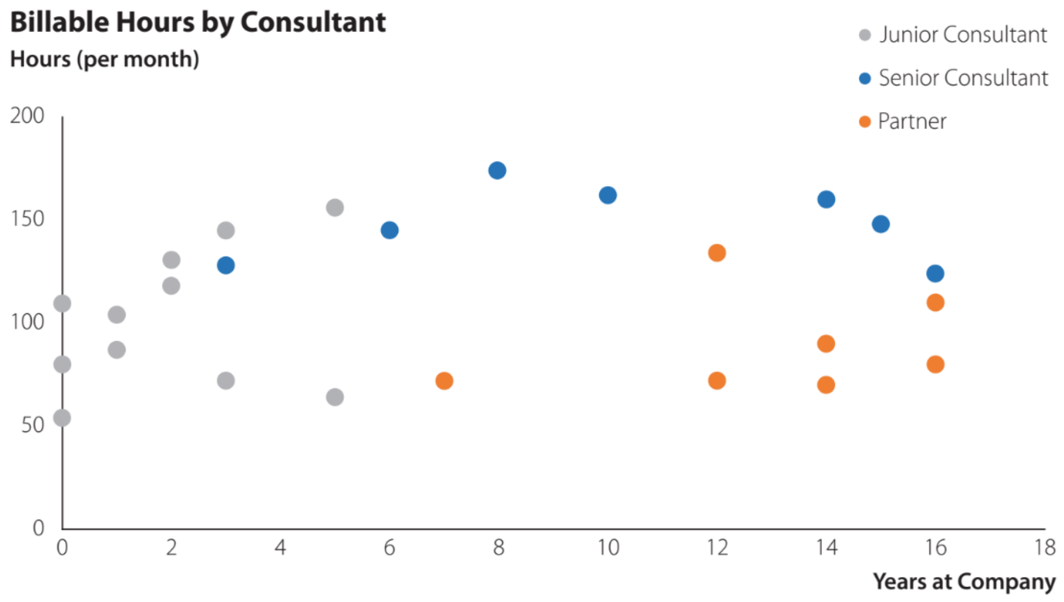
Too Many Attributes

Chart displaying billable hours association with years at the company, job title, and highest degree attained



Simplified chart displaying billable hours association with years at the company and job title



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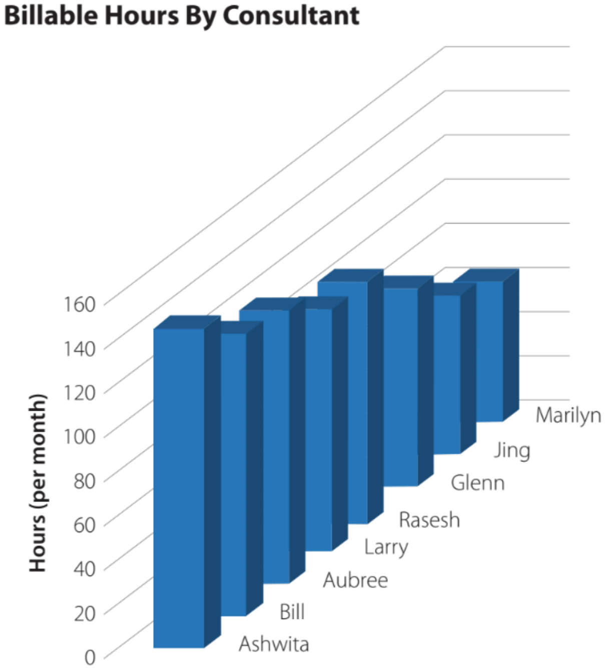
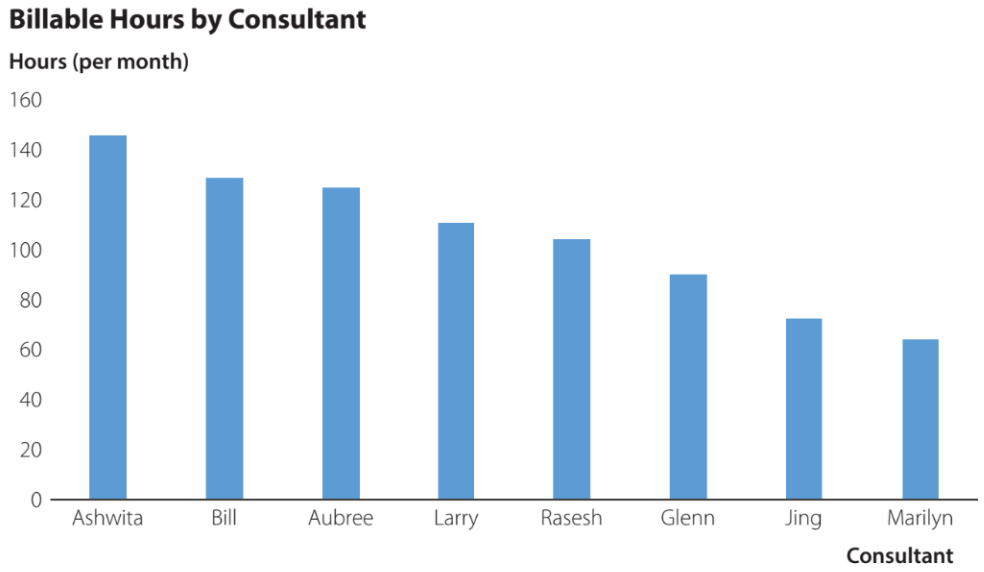
Unnecessary Use of 3D

A 3D column chart displaying billable hours – the third dimension is not useful



An improved 2D column chart displaying billable hours



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Summary

In this module you have learnt that:

- preattentive attributes direct attention
- the Gestalt principles help the audience to interpret objects
- data visualization strategies can help to:
  - increase the data-ink ratio
  - minimize eye travel for the audience
  - choose an effective font for text in data visualization.
- to avoid common mistakes in data visualization

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End of Part 3

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